

SHING TAK LAM

Mathematics Institute, University of Warwick
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EDUCATION

University of Warwick <i>Doctor of Philosophy (in progress)</i> PhD in Complex Differential Geometry supervised by Ruadhaí Dervan.	2025–28
University of Glasgow First year of PhD in Complex Differential Geometry supervised by Ruadhaí Dervan. Aligned student on the Algebra, Geometry and Quantum Fields CDT.	2024–25
University of Cambridge <i>Master of Mathematics</i> Courses in Algebra, Algebraic Geometry, Differential Geometry and Partial Differential Equations. Part III essay titled “Quantum cohomology and the Seidel representation” supervised by Jack Smith.	2023–24 <i>Distinction - 90%, ranked 17 out of 294</i>
University of Cambridge <i>Bachelor of Arts</i>	2020–23 <i>First class honours</i>

PREPRINTS

<i>Canonical Metrics on Families of Vector Bundles</i>	<i>To appear</i>
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CONFERENCE TALKS

Fibrations and Deformations , Brest <i>Canonical Metrics on Families of Vector Bundles</i>	<i>Jun 2025</i>
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SEMINAR TALKS

Warwick Postgraduate Seminar <i>Symmetries and Dynamics - Introduction to symplectic geometry</i>	<i>Nov 2025</i>
Warwick Junior Algebraic Geometry Seminar , <i>Canonical Metrics on Families of Vector Bundles</i>	<i>May 2025</i>

OTHER TALKS

AGQ CDT Reading group on Gauge theory <i>Compactness and Wall-Crossing</i>	<i>Nov 2025</i>
Learning seminar on <i>Moduli of weighted hyperplane arrangements</i> <i>Introduction to toric varieties</i>	<i>Nov 2025</i>
AGQ CDT Reading group on Gauge theory <i>Introduction to Gauge Theory and Seiberg–Witten invariants</i>	<i>Oct 2025</i>
Reading group on <i>K-stability of Fano Varieties</i> <i>α- and δ-invariants</i>	<i>Jun 2025</i>
Glasgow Geometry and Topology Pre-Seminar <i>Gauge theory and Floer theory</i>	<i>Mar 2025</i>
Reading group on <i>Positivity in Algebraic Geometry</i> <i>Big line bundles, Volume and Zariski decomposition</i>	<i>Feb 2025</i>
AGQ CDT Reading group on <i>Algebraic Stacks</i> <i>Deformation theory and smoothness</i>	<i>Feb 2025</i>
AGQ CDT Reading group on <i>Geometric Invariant Theory</i> <i>Kempf–Ness</i>	<i>Dec 2024</i>

AGQ CDT Examples showcase <i>Blowing up and rational elliptic surfaces</i>	<i>Nov 2024</i>
Part III Seminar <i>Symplectic Reduction</i>	<i>Mar 2024</i>
Part III Seminar <i>Morse theory</i>	<i>Dec 2023</i>
Cambridge Summer Research Festival <i>Hyperkähler structures on nilpotent $SL(n, \mathbb{C})$ orbits</i>	<i>Oct 2023</i>
Cambridge Summer Research Festival <i>On generalised hardness of approximation in optimisation and data science</i>	<i>Oct 2022</i>

ORGANISATION

Reading group on Geometric Invariant Theory	<i>Autumn 2024</i>
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EXPERIENCE

Summer research internship , University of Cambridge Summer project supervised by Alexei Kovalev, studying hyperkähler metrics on coadjoint orbits.	<i>Summer 2023</i>
Summer research internship , University of Cambridge Summer project supervised by Anders Hansen, studying (non-)computability of optimisation problems.	<i>Summer 2022</i>

AWARDS AND SCHOLARSHIPS

Clemmow/B Talbot Prize, K Roth Prize <i>Awarded for highest first in Part II Mathematics at Peterhouse</i>	<i>2023</i>
Senior Burkhill Prize for Mathematics, Routh prize for Mathematics <i>Awarded for highest first in Part IB Mathematics at Peterhouse</i>	<i>2022</i>
Summer Research Bursary <i>Funding from the Faculty of Mathematics for summer research projects.</i>	<i>2022, 2023</i>
Scholarship in Mathematics <i>Scholar of Peterhouse, University of Cambridge</i>	<i>2021-24</i>
College prize <i>College prize for academic achievement</i>	<i>2021, 2022, 2023, 2024</i>

CONFERENCES ATTENDED

VII BrAG Meeting , Nottingham	<i>Sep 2025</i>
Singular canonical Kähler metrics on compact and non-compact manifolds , Budapest	<i>Sep 2025</i>
Summer Research Institute in Algebraic Geometry , Colorado	<i>Jul 2025</i>
Fibrations and Deformations , Brest	<i>Jun 2025</i>
Winter School on K-stability , CIRM, Luminy	<i>Mar 2025</i>
UKAG Network Winter School , Lancaster	<i>Dec 2024</i>
GLEN , Glasgow	<i>Dec 2024</i>

SKILLS

Programming Languages Haskell, Python, Julia, Mathematica, Rust
Languages Native: English, Cantonese, Mandarin Basic: French, Spanish